Modeling Agent Customization and Optimization: A Category Theoretic Framework for Multi-Agent Workflows

J. Michael Constans and Gemini Pro 3.1.7

***Abstract*—As large language model (LLM) agents become increasingly modular, the orchestration of their components—such as discrete skills, model providers, and external data servers—presents a complex optimization problem. In this paper, we introduce a category theoretic framework for constructing and optimizing complex agent workflows. We define a strict monoidal category where objects are typed data streams and morphisms represent parameterized agent capabilities. By implementing this framework in Haskell using Generalized Algebraic Data Types (GADTs), we provide compile-time guarantees of well-formedness for agent topologies. Furthermore, we formalize the search for optimal agent configurations as a Bayesian Optimization problem over this categorical space, employing Pareto optimization to balance resource constraints against solution quality.**

***Index Terms*—Applied Category Theory, Large Language Models, Bayesian Optimization, Pareto Optimization, Monoidal Categories, Haskell.**

I. Introduction

The proliferation of agentic frameworks has enabled the construction of complex systems where multiple language models and tools interact to solve graduated tasks. However, discovering the optimal combination of these components under finite resource constraints (e.g., token limits and API latency) remains an open challenge.

In this work, we cast the formulation of agent architectures into the language of Applied Category Theory. Drawing inspiration from Waites' plumbing calculus for agent coordination, we define a copy-discard category that enforces structural boundaries on data flows. We then layer a Sequential Decision-Making framework over this category to iteratively evaluate and optimize these structures using Bayesian Optimization (BO).

II. Categorical Semantics of Agent Workflows

To ensure structural integrity, we define a category $\mathcal{C}$ where:

- **Objects** ($\mathrm{Ob}(\mathcal{C})$) are specific types of information states, such as Prompt, Context, Code, and TestResult.
- **Morphisms** ($\mathrm{Hom}(A, B)$) are computational processes mapping an input state to an output state.

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Gemini Pro 3.1 is a large language model developed by Google (model identifier `gemini-3.1-pro-preview`). This paper was prepared through interactive collaboration with that model: the human author directed the scope, theorems, and exposition; the model drafted prose, formalised statements, and produced the `tikz-cd` diagrams under continuous human review.

In our Haskell implementation, this category is realized via the `AgentGraph a b` Generalized Algebraic Data Type (GADT).

A. Monoidal Structure and Routing

Agents rarely operate in a purely linear sequence. To support parallelism and scatter-gather patterns, $\mathcal{C}$ is endowed with a strict monoidal structure, expressed via the tensor product $\otimes$ (implemented as `Par` or `***` in Haskell).

Furthermore, our category requires explicit structural routing morphisms, making it a copy-discard category:

$$A \xrightarrow{\Delta} A \otimes A \qquad (1)$$

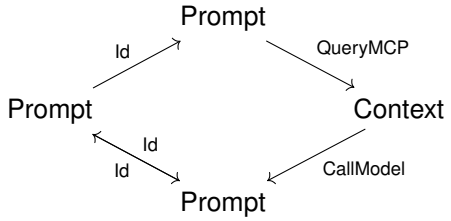
$$A \xrightarrow{\Diamond} I$$

Here, Δ (Copy) duplicates a stream, and $\Diamond$ (Drop) discards it, terminating the flow.

B. Composition of Morphisms

A complex agent workflow is built by composing simpler morphisms. For instance, consider an agent that duplicates a user prompt, uses one copy to fetch external context via a Model Context Protocol (MCP) server, and passes both the prompt and the new context to a generative model.

This topology is represented by the following commutative diagram:



(2)

In the DSL, this exact structure is guaranteed by the compiler:

```
standardPiAgent :: AgentGraph Prompt TestResult
standardPiAgent =
    Copy
    >>> (Id *** QueryMCP "github-pr-server")
    >>> CallModel "claude-3-7-sonnet"
    >>> ApplySkill "linter-skill"
    >>> RunTests
```

C. Advanced Control: Choices, Loops, and Parameterization

While directed acyclic topologies cover basic use cases, robust autonomous agents require discrete routing decisions and feedback loops (e.g., self-correction). We extend $\mathcal{C}$ with *Coproducts* (sum types) to model discrete conditional branching (`ArrowChoice`), and *Traces* to model feedback loops (`ArrowLoop`).

A feedback loop where a component D is fed back to the input is modeled categorically via a trace operator:

$$\begin{array}{ccc} B & \xrightarrow{\text{loop}} & C \\ {\scriptstyle \text{id}_B \otimes \text{trace}}\uparrow & & \uparrow{\scriptstyle \text{id}_C \otimes \text{trace}} \\ B \otimes D & \xrightarrow{f} & C \otimes D \end{array} \tag{3}$$

Furthermore, to optimize an agent, we must separate the *hyperparameter space* (e.g., model temperature) from the *data flow* (the prompt). Following Hanks et al. [3], we construct the parameterized category $\mathsf{Para}(\mathcal{C})$. A parameterized morphism is one where the computational process is contingent on an external parameter configuration P:

$$f : P \otimes A \to B \tag{4}$$

In our framework, this allows the Bayesian Optimizer to iteratively tune the parameter space P without altering the underlying topological validity of the graph from $A \to B$.

III. Case Studies in Complex Agent Modeling

To demonstrate the expressiveness of the $\mathcal{C}$ category and its Haskell implementation, we mapped two real-world, complex agentic architectures to our framework.

A. Claude Code: Orchestration and Dreaming

The March 2026 Anthropic Claude Code leak revealed a complex "lead-agent-plus-subagents" pattern, where a primary coordinator spawns isolated sub-agents in parallel. In our DSL, this is natively modeled using the monoidal tensor product $\otimes$ alongside Δ (Copy) to duplicate the context safely.

Furthermore, the leak highlighted a "Dreaming" routine for memory consolidation, which periodically compacts older conversation context. Because this involves taking a modified context and feeding it back into the agent loop, it cannot be modeled by a simple DAG. We use the trace operator (`ArrowLoop`) to mathematically model this cyclic dependency without deadlock.

B. OpenClaw: Embedded Runtimes and Event Taps

OpenClaw is an open-source workflow implemented directly on top of the Pi agent harness. Instead of executing the agent as an isolated subprocess, OpenClaw imports the agent session natively to inject custom tool pipelines and subscribe to real-time events for block chunking.

Our framework models this embedded paradigm elegantly. The custom policy filtering is modeled as a sequence of context-modifying skills. More importantly, the event subscription is modeled mathematically as a parallel process tapping the execution stream. We use Δ to copy the data stream, run the agent loop on the primary branch, and run the event subscriber on the secondary branch. The subscriber's output is then explicitly discarded via $\Diamond$ (Drop), ensuring the stream's structural integrity is maintained.

IV. Optimization Over Categories

Once the structural space of valid agent graphs is defined, we address the challenge of finding the optimal graph configuration. We frame this as a combinatorial Bayesian Optimization problem.

A. Pareto Frontier Evaluation

Each execution of a specific `AgentGraph` yields a set of `Metrics`, mapping to a multidimensional objective space (e.g., Token Cost vs. Quality). Because we cannot exhaustively evaluate all valid graphs, we rely on Pareto optimization.

Let $\Phi \in \Sigma_{\mathcal{C}}(S)$ represent a specific agent wiring. A configuration Φ_1 Pareto-dominates Φ_2 if it performs strictly better on at least one metric without degrading any others. The optimization seeks the Pareto Frontier—the set of non-dominated morphisms.

B. Surrogate Modeling and Acquisition

In continuous Bayesian Optimization, a Gaussian Process (GP) is often used as a surrogate model. Because our search space consists of discrete categorical diagrams, we utilize a graph-based surrogate model to predict the performance of untested configurations.

$$\mathsf{predictPerformance} : \mathsf{History} \to \mathsf{Hom}(A, B) \to \mathbb{E}[\mathsf{Metrics}] \tag{5}$$

The acquisition function systematically generates new morphisms (graphs) to test, explicitly balancing the exploration of novel agent topological structures against the exploitation of known, high-performing configurations.

V. Conclusion

By lifting agent orchestration into a strictly typed monoidal category, we decouple the structural validation of multi-agent systems from their execution. This abstraction provides a rigorous, mathematically sound foundation for deploying Sequential Decision-Making and Bayesian Optimization algorithms to autonomously design and refine the next generation of software engineering agents.

References

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